

HSE Intelligence — How the Web Console and Mobile Apps Exchange Data

What each mobile role produces, where it appears on the web console, and what the web console sends back down to the field. Role by role, in both directions.

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Workflows per ALGO/SMS-WF-2026-v1.0 · data classes per AI/HSMS Ver 1.0

ONE DATABASE

NO SYNC LAYER

The principle

The web console and the four mobile apps are **two views of one system**, not two systems that talk to each other. They share a single backend and a single database. Neither holds its own copy of the data.

This matters practically: there is **no synchronisation layer to build, monitor or reconcile**, and no possibility of the web console and a phone disagreeing about the same record. When a supervisor closes an incident on site, the manager's dashboard is already correct — nothing has to be pushed, exported or refreshed.

Configuration flows down

The web console defines how work should be done. Mobile enforces it as pickers, validation and gates.

Evidence flows up

Mobile captures what actually happened. The web console turns it into dashboards, KPIs and the audit trail.

Decisions run in the middle

Gate checks and score calculations happen on the server, so both sides always see the same verdict.

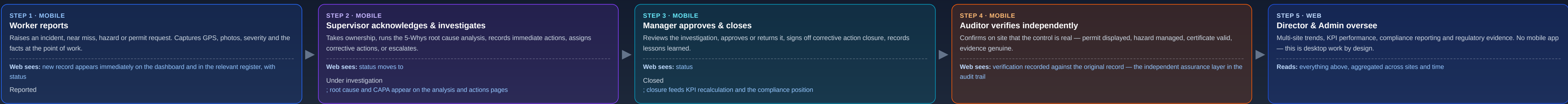
Configuration flows down · Web console → Mobile roles

What an administrator or Safety Manager sets up on the web becomes the rules the field app enforces

Org Setup · Sites & Zones Sites, departments, work areas, stations	Site hierarchy and every physical location where work happens → becomes the location picker on every mobile form. A station that does not exist on web cannot be selected in the field.	ALL FOUR ROLES Incident, near miss, risk, permit, safety walk, audit — every record is tagged to a real location
Users · People · Roles Employees, hierarchy, role assignment	Who exists, who reports to whom, and what each role may do → drives mobile login routing and every assignee picker. The backend role is authoritative; the app follows it.	ALL Supervisor Manager Team lists, CAPA owner selection, escalation targets, permit approver routing
Permit & Hazard Libraries Permit types, hazard categories, risk matrix	The catalogue of permit types and hazard classifications → becomes the selectable options when a worker raises a permit or reports a hazard, and sets which control checklist appears.	Worker Supervisor Manager Permit request form, hazard reporting, risk scoring bands
Policies · SOPs · Checklists Documents and checklist templates	Approved procedures and inspection templates → the checklist a worker completes and the audit template an auditor executes . Change the template on web, the next field execution uses it.	Worker Auditor Safety checklist, audit checklist execution, evidence requirements
Competence Matrix NEW Role → required course → validity period	Which qualifications each role must hold and for how long → becomes the worker's competence card , the supervisor's gap report , and the hard block that stops a permit when a safety-critical certificate has expired.	ALL FOUR ROLES Competence card · team matrix · permit gate · auditor evidence check
Threshold & Lookup Config NEW Score weights, limits, benchmark tables	Fatigue thresholds, performance-score weights and bands, weather operating limits by transport mode, corrective-action lookup tables and industry benchmarks → the numeric parameters every mobile gate and score uses . Tuned centrally, applied everywhere.	Worker Supervisor Manager Fatigue banding, journey go/no-go, score thresholds, alert triggers
Contractor Registry NEW Pre-qualification status and approved sites	Which contractor companies are approved, conditionally approved or barred, and for which sites → determines whether a contractor permit can be issued at all, and which contractor workers may be admitted.	Supervisor Manager Auditor Site access control, contractor permit gate, on-site verification
Assignments & Scheduling Audits, tasks, CAPA owners, shift patterns	Work allocated centrally → lands directly in the right person's mobile queue . An audit scheduled on web appears on the auditor's phone; a CAPA assigned on web appears in the owner's task list.	ALL FOUR ROLES Assigned audits, task list, CAPA queue, shift check-in options

THE WORKFLOW CHAIN · HOW A RECORD MOVES BETWEEN MOBILE ROLES — AND WHAT THE WEB CONSOLE SEES AT EACH STEP

Every workflow follows the same escalation path. The web console is not waiting at the end of the chain; it reflects each state change the moment it happens.



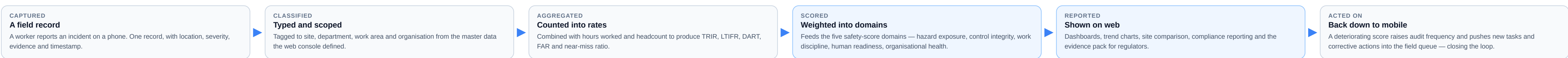
Evidence flows up · Mobile roles → Web console

What each role captures in the field, and the exact web module it lands in

WORKER Incidents · Near miss · Unsafe acts With GPS, photos, injury detail	Raw event records with severity, location, injured party, immediate cause and evidence photos → become the source of every incident statistic on the platform.	WEB Violations page · Near Miss page · Dashboard counters · Root Cause Analysis · Incident KPIs
WORKER Hazards · Risk reports · Permits Severity × probability, controls	Field-identified hazards and permit requests → populate the live risk register and the permit pipeline. Risk scores here drive the heat map on web.	WEB Risk page · hazard register · permit register · risk heat map
WORKER Checklists · Shift check-in · Fatigue NEW Completion, hours, rest	Checklist submissions and shift hours → evidence of routine compliance, and the raw input to the fatigue calculation that gates high-risk work.	WEB Checklist page · compliance evidence · Human Readiness domain of the safety score
SUPERVISOR Investigations · Root cause · CAPA 6-Whys, actions, owners, dates	The investigative substance behind each incident → becomes the root cause analysis record and the corrective action register that compliance reporting depends on.	WEB Root Cause Analysis page · Actions page · CAPA ageing · closure rate KPI
SUPERVISOR Safety walks · Team & shift · Toolbox talks Ratings, attendance	Proactive inspection and communication records → the leading indicators that show safety effort before an incident occurs, not after.	WEB Dashboard compliance ratings · leading indicator panel · communication coverage
SUPERVISOR RAMS scores · Gate overrides NEW Rubric score, override rationale	Contractor method-statement scoring, and every decision to proceed past a fatigue or competence warning with a written reason → the regulatory defensibility record .	WEB Contractor scorecard · override audit log · Work Discipline domain of the safety score
SAFETY MANAGER Approvals · Closures · Sign-off Permits, incidents, CAPA, hazards	The authority decisions that close each workflow → determine what counts as resolved in every KPI, and complete the compliance chain of custody.	WEB Dashboard · Analytics · compliance status · CAPA closure rate · permit register
AUDITOR Audit findings · Verifications Pass / fail, scores, evidence	Independent assurance — checklist results, compliance scores, and on-site confirmation that permits, hazards and certificates are genuine → the evidence an external regulator will ask for .	WEB Compliance page · audit register · compliance score · audit trail · regulatory evidence pack
WORKER Supervisor Journey plans · Check-ins NEW Route, vehicle, timed check-ins	Journey risk scores, pre-trip inspections and timed check-ins → the transport safety position , including missed check-ins that trigger escalation.	WEB Fleet dashboard · journey risk distribution · vehicle defect rate · transport KPIs

FROM A FIELD RECORD TO A BOARD-LEVEL NUMBER

How individual mobile entries become the KPIs and scores the web console reports. Nothing is typed in twice; every figure traces back to a field record.



INTERACTION MATRIX · WHO TOUCHES WHICH DATA

Read across a row to see how one data domain is handled by the web console and by each mobile role.

DATA DOMAIN	WEB CONSOLE	WORKER	SUPERVISOR	SAFETY MANAGER	AUDITOR	WHERE THE AUTHORITY SITS
MASTER DATA — defined centrally, consumed in the field						
Sites, work areas, stations	Creates & maintains	READ	READ	READ	READ	Web — System Admin
Employees, roles, hierarchy	Creates & maintains	READ	READ	READ	READ	Web — System Admin
Permit types, hazard libraries	Creates & maintains	READ	READ	READ	READ	Web — Safety Manager
Competence matrix NEW	Creates & maintains	READ	READ	READ • WRITE	READ	Web config · Manager may amend on mobile
Thresholds & lookup tables NEW	Creates & maintains	—	READ	READ	READ	Web only — deliberately not field-editable
OPERATIONAL DATA — captured in the field, reported centrally						
Incidents, near misses, unsafe acts	Reads & reports	CREATES	INVESTIGATES	CLOSES	READ	Mobile — full lifecycle
Permits to work	Reads & reports	REQUESTS	ACKNOWLEDGES	APPROVES	VERIFIES	Mobile — all four roles participate
Hazards & risk register	Reads · heat map	REPORTS	READ	REVIEWS	VERIFIES	Shared — mobile captures, web analyses
Corrective actions (CAPA)	Reads · ageing report	OWNS TASK	ASSIGNS	SIGNS OFF	READ	Mobile — web reports on it
Audits & inspections	Schedules · scores	—	SAFETY WALK	READ	EXECUTES	Web schedules · mobile executes
Training & competence records NEW	Matrix & effectiveness	COMPLETES	VERIFIES TEAM	APPROVES	AUDITS	Shared — all four roles participate
Fatigue & shift hours NEW	Reads · flag rate	DECLARES	OVERRIDES	AUTHORISES EXCEPTION	AUDITS	Mobile — escalating authority
Contractor scoring NEW	Registry & scorecard	—	SCORES RAMS	PRE-QUALIFIES	AUDITS	Web holds registry · mobile scores
Journey plans & check-ins NEW	Fleet KPIs	PLANS & CHECKS IN	AUTHORISES	OVERSEES	AUDITS	Mobile — inherently a field activity
DERIVED DATA — calculated on the server from everything above						
Safety KPIs — TRIR, LTIFR, DART, FAR	Primary consumer	—	TEAM VIEW	SITE VIEW	READ	Server-calculated — no manual entry anywhere
Safety performance score NEW	Configures & reports	OWN SCORE	TEAM	SITE	VERIFIES DATA	Server-calculated · web sets the weights
Gate decisions & override log NEW	Audit trail	SEES VERDICT	OVERRIDES	EXCEPTIONS	AUDITS	Server decides · both clients display

DEPENDENCIES WORTH KNOWING

Places where the two sides rely on each other, and what happens if one is not in place.

Master data must exist before field capture A mobile form can only offer locations, permit types and hazard categories that exist on web. An incomplete site setup shows up immediately as empty pickers in the field.	The competence matrix gates the permit Until the matrix is populated on web, the mobile permit gate has nothing to check against. Training configuration is therefore a prerequisite for permit control , not a parallel task.	Thresholds are deliberately web-only Fatigue limits, score weights and weather limits cannot be edited from a phone. This is intentional — a safety limit should not be changeable by the person it constrains .	KPIs are only as good as field capture Every rate on the web dashboard divides field-reported events by exposure hours. Gaps in mobile reporting appear as improvements in the KPI, which is why data-completeness checking matters .
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Web-authored Mobile-captured Creates the record Reads and writes

One backend, one database, no synchronisation layer. Every figure on the web console traces back to a record someone captured in the field, and every rule the field app enforces was set on the web console.